

Paying More Did Not Bring More Power

Edition 3 in the Energy, Infrastructure & Climate Resilience series. Previous Decision Intelligence briefing: 'The Summer the Grid Said No' (July 2026).

Cycle: 17 September 2026 · Audience: Senior strategy, foresight, infrastructure, risk and policy professionals · Horizon: to 2030

Read estimate: 2 min Executive Synthesis · 29 min full read

What you will learn

1. Why a power price stuck at its legal ceiling now means supply is missing, not that it is on the way
2. How four regulators turned a request to connect to the grid into a long contract with conditions, deposits and exit costs
3. What will actually hold back your 2029 plan: waiting times for equipment, the terms of connecting, or cooling water

Key takeaways

1. PJM's 2028/2029 auction hit its price cap of \$325.00/MW-day, finished 6,831.3 MW short, and bought just 317.1 MW of new generation
2. Twenty-three US states now have special tariffs for very large electricity users; Oregon's run up to 30 years, with minimum charges at 90% of reserved capacity
3. Romania's Cernavoda nuclear plant went fully offline for only the second time ever, and low water on the Rhine cut coal barge loadings by three-quarters

Why This Matters Now

Electricity buyers spent most of 2026 discovering that money is no longer the binding constraint. Prices hit their ceilings, bills went up, and almost no new power arrived, because what is actually scarce is the equipment, the permission and the water to run a plant, not the willingness to pay for one.

Three responses are under way at once. Regulators on three continents have rewritten what a large new customer must sign before it can connect. Governments have started deciding in public who carries the cost of the network that customer triggers. And the manufacturers have sold out their factories to 2030.

The one thing to take into the next meeting: treat a connection as a long contract with obligations, not as a place in a queue, and check whether anything you own or finance depends on a river behaving the way it used to. The early warning is in the terms of the contract, not the price.

Executive Synthesis

Which decisions on your desk this week assume that a high enough price will bring the power?

The evidence in this report says the highest price the rules allow bought 317 megawatts.

This cycle the electricity system reached the top of its price signal and got almost nothing back. PJM, which runs the grid for 13 US states and Washington DC, held its auction for power in 2028/2029: it cleared at the highest price its rules allow, finished 6,831.3 MW short of what reliability requires, and brought forward 317.1 MW of new generation ([PJM 2028/2029 Base Residual Auction Report](#), Jul 2026). Money is no longer what is scarce. Equipment, permission to connect and cooling water are.

Three terms are being rewritten at once. Regulators in the United States, Britain and Japan have turned a request to connect a large new customer, usually a data centre, into a long contract with conditions ([McGuireWoods](#), Jun 2026) ([Ofgem](#), Jul 2026). Governments are deciding in public who pays for the network that customer needs: twenty-three US states now have a special tariff for large loads ([Columbia's Sabin Center](#), Jun 2026). And the turbine makers have sold their output to the end of the decade ([GE Vernova](#), Jul 2026). A dry summer added a warning of its own, as low and warm rivers took nuclear and coal plants offline in Romania, France and along the Rhine ([Euronews](#), Aug 2026) ([Insurance Journal](#), Jul 2026).

Looking ahead to 2030, the IEA expects data centres worldwide to use about 950 TWh of electricity, roughly double 2025, with 15 to 27 GW of it possibly supplied by gas plants built on site ([IEA](#), Apr 2026). Developers are chasing 63 GW of new gas capacity for

2026 to 2030 ([Wood Mackenzie, reported by Utility Dive](#), Apr 2026), while GE Vernova plans to reach 30 GW of annual output only by 2030 ([GE Vernova](#), Jul 2026). None of this is certain, and each is a place where readers can still choose their position: section 7 sets out the options.

Where this analysis could be wrong

The analysis assumes the shortfall is a delivery problem, not a forecasting one. If the large loads being planned for do not arrive, the auctions are pricing demand that never comes: the Texas grid's queue holds around 460,000 MW of interest against a system that peaked at 91.3 GW ([The Texas Tribune](#), Jul 2026). The warning sign would be the largest cloud companies cutting their spending plans for two quarters running. So far spending is still rising, but more of it is borrowed: debt now funds 32% of it, against 9% two years earlier ([FactSet](#), Jul 2026).

The decisions that cannot be deferred this cycle

- 1. **Treat every large new connection as a long contract, not a place in a queue.** Oregon's approved terms run up to thirty years, with minimum charges set at 90% of the capacity the customer contracts.
- 2. **Secure a delivery slot for equipment rather than a choice of technology.** With the turbine makers sold out to 2030 and large transformers taking four years, the manufacturing window is the scarce asset.
- 3. **Test every river-cooled plant and barge-supplied fuel contract against a dry summer.** The Rhine and the Danube took generation and fuel deliveries offline this summer.

Each is developed below, with a decision posture, in the four Strategic Implications, and with alternatives in the Options.

Continuity ledger: what the July edition said and what happened

Seven items: the six milestones the 16 July edition set for this cycle, plus one correction to a figure that edition published. Each carries a verdict and a post-cycle source.

Obligation	Verdict	What changed
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The price in PJM's 2028/2029 capacity auction

CONFIRMED

Cleared at the \$325.00/MW-day administrative cap, 6,831.3 MW short of requirement, with 317.1 MW of new generation ([PJM 2028/2029 Base Residual Auction Report](#), Jul 2026).

Whether any grid actually orders data centres to cut their power use

BROKEN

It did not. PJM kept the emergency power it was given on 18 May ([Utility Dive](#), May 2026), met an all-time peak of 168,158 MW on 2 July with about 6,113 MW of customers cutting their use on request, and says it never had to order data centres onto their backup generators ([PJM](#), Jul 2026).

Correction to the 16 July edition

REOPENED

That edition reported the 2 July peak as roughly 162.7 GW, below the 2006 record. PJM's own figure is 168,158 MW, which broke the 165,563 MW record ([PJM](#), Jul 2026). The July reading that the record held is withdrawn.

Spending by the largest cloud companies in the second quarter of 2026

CONFIRMED

Capex held and rose, reaching roughly \$490bn across five hyperscalers in the twelve months to May 2026, but incremental debt as a share of capex went from 9% to 32% ([FactSet](#), Jul 2026): the buildout confirmed, the funding changed.

The 2026 Atlantic hurricane season against the bet that reinsurance prices would keep falling

CONFIRMED

First-half insured catastrophe losses were \$46bn, the lowest since 2019 and 28% below the ten-year average, and the price insurers pay to insure their own property risks fell 12% in January and 16% in July ([Insurance Business](#), Jul 2026). The season itself is not over.

Whether refused water permits start to decide where data centres can be built

CONFIRMED

No headline denial, but Virginia's environmental regulator found no location east of Interstate 95 can support a three million gallon per day industrial groundwater withdrawal under current standards ([Data Center Knowledge](#), Jul 2026), which is a stronger gate than a refusal.

Growth of California's insurer of last resort, the FAIR Plan

UNRESOLVED

Growth decelerated, at 15,779 policies and \$27bn added in the first quarter, still 25% higher year on year ([Intelligent Insurer](#), Apr 2026). The California wildfire-credit theme is not carried this cycle, so this milestone is handed forward rather than closed.

Results of the Bank of
England's climate
scenario and insurance
stress test

UNRESOLVED

The test itself finished in May 2026, and the Bank says it will publish overall findings later in 2026 ([the Bank of England](#)). Carried to the next cycle.

Audience Snapshots

Four cards, one per reader group. Each says what is changing for that group and suggests one question for the next meeting, so a team can take its card as an agenda item without reading the whole report.

IF YOU ARE TRYING TO CONNECT A LARGE
NEW LOAD

Connecting to the grid now means signing a long contract with conditions

What is changing: Oregon's approved terms run 10 to 30 years, with the customer paying all distribution upgrade costs ([Utility Dive](#), May 2026), and Texas requires a large load to be able to disconnect fully within thirty minutes ([White & Case](#), Aug 2026).

Question for your next meeting: Does our site still pay off with a thirty-year minimum charge at 90% of contracted capacity, and who inside the business owns that liability?

What it is for: Turns each connection into a liability that a named person owns, before the next capital approval.

IF YOU SET OR APPROVE NETWORK TARIFFS

Four jurisdictions have chosen four different ways to share the cost of the grid

What is changing: Twenty-three US states now have a special tariff for large loads ([Columbia's Sabin Center](#), Jun 2026), Spain would make network access itself revocable ([Aragon Digital](#), Sep 2026), and Germany is subsidising transmission charges to hold bills down ([Bundesnetzagentur](#), May 2026).

Question for your next meeting: If the forecast load does not arrive, who pays for the network built for it: the developer or the household?

What it is for: Shows whether a tariff protects households when demand disappoints, not only when it arrives.

IF YOU FINANCE OR INSURE GENERATION ASSETS

The scarce asset is a delivery slot, not the plant

What is changing: Turbine order books are full to the end of the decade ([GE Vernova](#), Jul 2026), new gas plants cost at least \$2,500 per kW ([IEEFA](#), Apr 2026), and what came online in the first half of 2026 was mostly solar and batteries ([Inside Climate News](#), Aug 2026).

Question for your next meeting: Does our capital plan assume a gas unit we cannot take delivery of before 2030, and what does it cost to bridge the gap with solar and storage?

What it is for: Separates the plans that depend on equipment nobody can deliver from those that do not.

IF YOUR PLANT DEPENDS ON RIVER OR SEA WATER FOR COOLING

Low rivers and warm water now take plants offline for weeks

What is changing: Romania's Cernavoda nuclear plant went fully offline ([Euronews](#), Aug 2026), a record 15.6% of French nuclear capacity was missing in July ([Agence France-Presse, via Boursorama](#), Aug 2026), and low water on the Rhine cut coal barge loadings by about three-quarters ([Insurance Journal](#), Jul 2026).

Question for your next meeting: What happens to our revenue and reliability if our water limit binds for six weeks rather than six days, and does any contract we hold price that?

What it is for: Puts a number on a risk that most valuation models leave out.

Themes

The cycle's signals are organised into five themes. The tier (**Immediate**, **Near-Term**, **Longer-Range**) says when a theme bites. The weight says how much it shapes readers' decisions to 2030, and the last two columns say who decides and how much room readers have to act. Weights are the analysis's judgement, stated so they can be challenged.

Theme	Weight	Who decides	Room to act
1. Paying more did not buy more power	HIGH	Grid operators such as PJM, and the state governments that negotiate their price caps	Low for most buyers, who pay the price set; medium for large buyers able to contract for supply directly

Theme	Weight	Who decides	Room to act
2. The regulator rewrote the rules for every large load	HIGH	Regulators: the US federal energy regulator, the Texas utility commission, Ofgem and Japan's energy agency	High for developers, who choose where to connect and on what terms
3. Who pays for the grid became the live political question	HIGH	State regulators and governments, often in an election year	Medium: firms can shape terms in consultations; households have little say
4. You cannot buy a turbine before 2030	MEDIUM	The three large turbine makers, and whoever already holds a delivery slot	High for buyers who can reserve a slot early or switch to solar, storage and flexible demand
5. Heat and low water stopped being a weather story	MEDIUM	Nobody: the weather decides, and plant operators and river authorities manage the consequences	High for asset owners, lenders and insurers, who can price the risk and adapt the assets

1. Paying more did not buy more power

IMMEDIATE

Weight: HIGH · **Who decides:** Grid operators such as PJM, and the state governments that negotiate their price caps · **Room to act:** Low for most buyers, who pay the price set; medium for large buyers able to contract for supply directly

The price your investment plan relies on has stopped bringing in new supply, and that changes what a charge for capacity tells you. PJM's 2028/2029 auction hit the price cap set by its rules, closed 6,831.3 MW below the amount it needs to keep the lights on, and brought forward only 317.1 MW of new generation ([PJM 2028/2029 Base Residual Auction Report](#), Jul 2026). Any plan that reads a rising price as a sign that supply is coming has it the wrong way round: it is a sign that supply is missing. The cost of the shortage now lands on household bills and in politics, rather than paying for new plants.

PJM 2028/2029 Base Residual Auction: what the ceiling price procured

Megawatts, 2028/2029 delivery year

Shortfall against the reliability requirement

6,831.3

317.1

New generation procured

Clearing price: **\$325.00/MW-day UCAP**, the administrative cap, for the third auction running.

New generation covered **4.6%** of the shortfall the price was meant to close.

Source: PJM 2028/2029 Base Residual Auction Report, July 2026. Bars are to scale against each other.

- The auction hit its price cap for the third time running, so the price no longer tells anyone how short of power the system is ([Utility Dive](#), Jul 2026).
- PJM says the auction is working as designed, not failing. Its paper on market design describes a trap in which the high prices needed to attract investment provoke political intervention that scares investment off, and states that “The market is generating the price signal it was designed to generate during a shortage.” ([PJM, Powering Reliability Through Market Design](#), May 2026)
- The cost is already in household bills. New Jersey regulators put the increase at around 20%, with capacity charges accounting for 15% to 20% of a residential bill ([New Jersey Board of Public Utilities](#), Aug 2026).
- Texas shows the contrast. Its grid operator, ERCOT, met a record 91.1 GW of demand in one hour on 22 July, 6% above the 2023 record, mainly with gas at 48% and solar at 32% ([US Energy Information Administration](#), Aug 2026), and still had more than 20,000 MW of spare capacity lined up when the record was set ([The Texas Tribune](#), Jul 2026).

Outlook to 2030

PJM itself sets out three ways to reform its market, including openly rationing power by place or by type of customer ([PJM, Powering Reliability Through Market Design](#), May 2026). In Texas, the grid operator is planning for demand of roughly 175,000 MW by 2032, almost double this summer's record ([The Texas Tribune](#), Jul 2026).

What it means: By 2030 the question is less what capacity costs than who gets it when there is not enough. Buyers who contract directly for supply now are less exposed to how that rationing is designed.

Counter-argument

The strongest challenge is that an auction with a price cap cannot be judged on what it bought, because the cap itself kept new supply out: a higher price might have attracted more, and the states that negotiated the cap own that choice. Sierra Club argues the opposite, that annual customer costs in PJM have risen by \$12.5bn since June 2025 and the cap is too loose rather than too tight ([Sierra Club](#), Jul 2026). Both readings can hold, and they converge on the same test: whether the 2029/2030 auction, run under whatever cap then applies, brings forward materially more than 317.1 MW. If it does, the cap was the problem. If it does not, the problem is getting new plants built, and this theme holds.

Decision link: Strategic Implications [1](#) and [4](#).

2. The regulator rewrote the rules for every large load

IMMEDIATE

Weight: HIGH · **Who decides:** Regulators: the US federal energy regulator, the Texas utility commission, Ofgem and Japan's energy agency · **Room to act:** High for developers, who choose where to connect and on what terms

Within four months, four jurisdictions each turned a request to connect a large customer from an engineering question into a contract with conditions, without coordinating with each other. If your growth plan assumes the terms you signed under, or expected to sign under, it is already out of date in at least one market you operate in. In practice, the question has moved from whether power is available to what you must accept in order to get it.

- On 18 June FERC, the US federal energy regulator, ordered all six regional grid operators to justify or change how they connect very large users, with a 17 August deadline to respond ([McGuireWoods](#), Jun 2026); it chose regional rules over a national standard and kept the option to step in later ([Utility Dive](#), Jun 2026).
- Texas went furthest on how connections work day to day: the state utility commission's 24 July order confirms that a large user sited next to a power plant must be able to disconnect fully within thirty minutes, whatever that plant is producing ([White & Case](#), Aug 2026).
- Britain went after the queue rather than the connection. Ofgem, the energy regulator, is clearing speculative projects from a queue that has grown from 41 GW to 125 GW, roughly 80 GW of it data centres, and is introducing a fee to show commitment ([Ofgem](#), Jul 2026), with deposits, fees to stay in the queue and deadlines after an offer is made all proposed for new users ([Herbert Smith Freehills Kramer](#), Jul 2026).
- Japan regulated the data centres themselves rather than their connection: from April 2026 operators must report their electricity use and their efficiency rating (PUE, where 1.0 is perfect), publish targets every year, and reach 1.4 or better by fiscal 2030, or 1.3 for centres built from fiscal 2029 ([Japan's Agency for Natural Resources and Energy](#), May 2026).

Early indicators

- **WEAK SIGNAL** **Losing the connection as the penalty.** Spain's draft decree would cut a data centre off from the grid if it repeatedly fails to get 80% of its power from new renewable sources. The Spanish is unambiguous: “El incumplimiento reiterado podrá acarrear la pérdida del acceso a la red.” ([Aragon Digital](#), Sep 2026) It gains weight if a second EU country copies the threat of disconnection rather than a fine.
- **WEAK SIGNAL** **Efficiency standards as a condition of entry.** Japan's requirement that centres built from fiscal 2029 reach an efficiency rating of 1.3 ([Japan's Agency for Natural Resources and Energy](#), May 2026) is a hard number aimed at the data centre rather than the grid. It gains weight if a US state or the European Commission writes a minimum efficiency into a tariff.

Outlook to 2030

The IEA expects data centres worldwide to roughly double their electricity use, from 485 TWh in 2025 to 950 TWh in 2030 ([IEA](#), Apr 2026). Japan's data centres must reach an efficiency rating of 1.4 or better by fiscal 2030 ([Japan's Agency for Natural Resources and Energy](#), May 2026).

What it means: Connection terms written now will govern a much larger load by 2030. Developers who take part in writing them now face fewer surprises later.

Counter-argument

The reforms may change the price of connecting without limiting who connects. FERC explicitly left states in charge of retail sales, who qualifies and where plants are built ([Utility Dive](#), Jun 2026), and its own rulemaking file describes the aim as connection that is “timely, orderly, reliable, and non-discriminatory” ([FERC Docket RM26-4](#), accessed Sep 2026), which is a push to connect faster as much as a restriction. NERC, which sets reliability standards for the North American grid, records 58.5 GW of new supply against 11 GW of extra peak demand ([NERC 2026 Summer Reliability Assessment](#), May 2026), so the case that there is not enough power is not the same everywhere. The test is whether any of the six grid operators' responses after 17 August actually reduces how much new demand can connect, rather than changing who connects first.

Decision link: Strategic Implications [1](#) and [4](#).

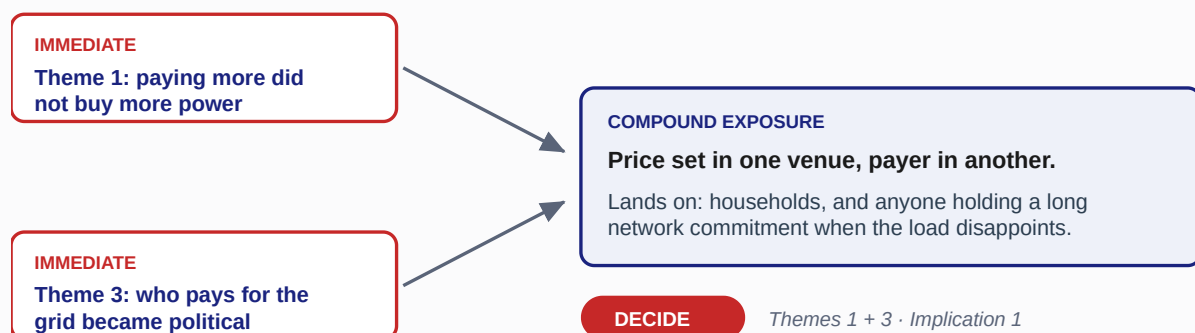
3. Who pays for the grid became the live political question

IMMEDIATE

Weight: HIGH · **Who decides:** State regulators and governments, often in an election year ·
Room to act: Medium: firms can shape terms in consultations; households have little say

Who pays for the grid stopped being a technical detail of utility pricing and became a real policy tool, decided in public and often in an election year. For anyone building, financing or supplying a very large user, the tool each jurisdiction chooses now changes the project's costs materially, and the tools differ more than the problem does. The same data centre carries different costs in Oregon, Madrid and Munich.

Where Theme 1 meets Theme 3



Drawn from the cross-impact grid in the Energy, Infrastructure and Climate Resilience Change Tracker (27 August 2026 cycle), re-sourced against this cycle's evidence base.

- Twenty-three US states have at least one special tariff for very large users and seven more have proposals pending, with Virginia's taking effect on 1 January 2027 ([Columbia's Sabin Center](#), Jun 2026); regulators approved 29 such tariffs in 2025 alone, against 14 in the whole of 2018 to 2024 ([Smart Electric Power Alliance](#), Apr 2026).
- Oregon shows what a settled tariff looks like: contracts of 10 to 30 years depending on size, the customer paying 100% of the cost of upgrading the local network, a minimum charge of 90% of the capacity it reserves, and a 1 cent/kWh surcharge above 100 MW passed back to households and low-income customers ([Utility Dive](#), May 2026).
- The federal government answered with a voluntary promise rather than a tariff: the expanded Ratepayer Protection Pledge claims to cover 80% of all power delivered to US homes and businesses ([the White House](#), Jul 2026), a claim that depends on those who signed it keeping their word.
- Europe is using two different levers at once: Germany has directed a federal subsidy at transmission charges and reformed network-cost sharing ([Bundesnetzagentur](#), May 2026), while the Commission proposed in July to cut network charges for specific consumer groups ([European Commission](#), updated Jul 2026) and Spain would attach a renewable-sourcing condition to access itself ([Aragon Digital](#), Sep 2026).

Outlook to 2030

Pennsylvania projects the price cap will be worth roughly \$45bn to 67 million people over four years ([Governor of Pennsylvania](#), Jul 2026). But the IEA sees 15 to 27 GW of gas plants built on site at data centres by 2030 ([IEA](#), Apr 2026), which takes load, and the customers who pay for the network, off the shared grid.

What it means: The more large loads supply themselves, the fewer customers are left to pay for the grid. By 2030 the argument may move from the cost of connecting to the cost of leaving.

Counter-argument

A tariff written for a boom is tested by a bust. If the expected demand does not arrive, households still pay for the network built for it, and terms that make the developer pay are only as good as the developer's finances in 2031. The clearest warning is in how the building is paid for rather than in the tariff: the share of the largest cloud companies' spending funded by new debt rose from 9% in financial year 2024 to 32% in the year to June 2026 ([FactSet](#), Jul 2026). On the available evidence the tariffs move the cost but not the risk that demand falls short, and whatever is left over returns to household bills by default.

Decision link: Strategic Implications [1](#), [3](#) and [4](#).

4. You cannot buy a turbine before 2030

NEAR-TERM

Weight: MEDIUM · **Who decides:** The three large turbine makers, and whoever already holds a delivery slot · **Room to act:** High for buyers who can reserve a slot early or switch to solar, storage and flexible demand

New gas plants are not just expensive, they are unavailable, and that upsets every investment plan built on them over the next twelve months. The three makers of large gas turbines have sold their output to the end of the decade, other equipment such as transformers takes as long, and prices have risen faster than utilities planned for. Whoever holds a delivery slot holds the scarce asset; everyone else must choose between waiting and building something else.

- GE Vernova's own quarterly results are the clearest measure of the gap: orders and reserved production slots for its gas turbines grew from 100 GW to 116 GW in one quarter, and it expects at least 125 GW by the end of 2026 ([GE Vernova](#), Jul 2026). It plans to make 30 GW a year by 2030, later than most of today's plans need it.

- Mitsubishi's order book for large turbines reached 35 GW against 23 GW a year earlier, and orders placed in the first quarter of 2026 are due for delivery between 2028 and 2030 ([Utility Dive](#), Aug 2026); Siemens Energy's stands near 69 GW with waiting times of more than three years ([Utility Dive](#), Aug 2026) and Ansaldo is expanding to fill the same gap ([POWER](#), Sep 2026).
- Other equipment is no better: large transformers now take four years to deliver, and demand for the transformers that connect power plants to the grid rose 274% between 2019 and 2025 ([pv magazine USA](#), May 2026).
- Prices have risen with scarcity. IEEFA puts the newest proposed gas plants at \$2,500 per kW installed or more, or \$2.5bn for a 1,000 MW plant ([IEEFA](#), Apr 2026), while utilities told investors in the second quarter that they are focused on delivering projects and protecting customers rather than negotiating on price ([Utility Dive](#), Aug 2026).

Outlook to 2030

Wood Mackenzie expects turbine prices to reach \$600/kW by the end of 2027, up 195% since 2019, as developers chase 63 GW of gas capacity for 2026 to 2030 ([Wood Mackenzie, reported by Utility Dive](#), Apr 2026). GE Vernova plans to raise its annual output to 30 GW only by 2030 ([GE Vernova](#), Jul 2026).

What it means: Gas ordered today mostly arrives around 2030. Until then, new supply comes from solar, batteries and demand that can flex, and plans should be built around them.

Counter-argument

A shortage of one technology is not a shortage of supply. In the first half of 2026 the United States brought online 11,458 MW of utility-scale solar and 8,207 MW of batteries against 2,707 MW of natural gas across 46 projects ([Inside Climate News](#), Aug 2026), so the shortage limits the gas route, not new capacity as a whole, and the manufacturers are doubling their output against 2024 levels ([GE Vernova](#), Jul 2026). The limit of that argument is the season: solar at 32% of a Texan evening peak ([US Energy Information Administration](#), Aug 2026) is not the same as power that is guaranteed on a still winter night, which is the hour the capacity auction is actually buying for.

Decision link: Strategic Implications [2](#) and [3](#).

5. Heat and low water stopped being a weather story

NEAR-TERM

Weight: MEDIUM · **Who decides:** Nobody: the weather decides, and plant operators and river authorities manage the consequences · **Room to act:** High for asset owners, lenders and insurers, who can price the risk and adapt the assets

How much water a river carries, and how warm it is, moved from a footnote in outage reports to a factor in which plants run, how fuel moves and what plants are worth, across two river basins in one season. If you own, finance or buy from a coal, gas or nuclear plant that takes cooling water from a river, your assumption about how often it can run is now the number most likely to be wrong. Heat also pushes demand up, which is why it belongs in the discussion about capacity rather than beside it.

- Romania's Cernavoda plant, normally a fifth of national electricity, was fully offline in August for only the second time in its history, the first being 2003 ([Euronews](#), Aug 2026); the water authority put Danube flow at Romania's border at 1,650 cubic metres per second against a July average of 4,750, approaching the 1985 minimum of 1,400 ([Mediafax](#), Jul 2026).
- France lost output through the summer: around 3% in June and around 6% in July, with a record 15.6% of capacity missing ([Agence France-Presse, via Boursorama](#), Aug 2026), and 13 of 57 reactors affected at the peak of the episode ([Selectra](#), Aug 2026).
- Fuel could not move either. The Rhine at Kaub fell to 28 centimetres, the lowest since 2018, with coal inventories at Amsterdam, Rotterdam and Antwerp up about 75% since spring while barge loadings inland were cut by three-quarters ([Insurance Journal](#), Jul 2026): the coal existed, it was simply in the wrong place.
- Alpine hydro ran at 51% of normal in Austria, 48% in Switzerland, 63% in Germany and 68% in France in July ([Euronews](#), Aug 2026), and gas filled the gap: EU gas generation rose 8% last year as hydro fell 12%.

Early indicators

- **WEAK SIGNAL** **Sea life as a cause of outages.** Jellyfish blockages at Gravelines sat alongside heat as a cause of lost French output this summer ([Agence France-Presse, via Boursorama](#), Aug 2026). It gains weight if a second operator reports sea or river life blocking its cooling-water intake in another region within a year.
- **WEAK SIGNAL** **A broader but shallower drought.** The Joint Research Centre records half of the EU and the UK in drought but only 9% of territory at alert level in July, against 18% at the 2022 peak ([the Commission's Joint Research Centre](#), Aug 2026). It points the other way if the 2027 season is again widespread but mild rather than intense.

Outlook to 2030

EDF's losses to heat and low water, around 0.3% of output a year, could reach 1.4% by 2035 ([Selectra](#), Aug 2026), and energy used for cooling worldwide is set to double by 2050 against 2023 ([MIT Technology Review](#), Jun 2026), so hot summers raise demand at the moment they cut supply.

What it means: By 2030 a dry summer is a planning case, not an exception. Plants and fuel contracts tested against one now are worth more than those that are not.

Counter-argument

2026 may have been widespread rather than severe. The Joint Research Centre's own comparison puts 9% of EU and UK territory at alert level in July against 18% at the 2022 peak and 6% in late August 2018 ([the Commission's Joint Research Centre](#), Aug 2026), which is a different shape of event rather than an unambiguously worse one. France's system operator stated before the summer that security of supply required no particular vigilance ([RTE](#), Jun 2026), and it was right: no shortfall materialised. So the theme claims a risk to how often plants can run and what that costs, not a risk of blackouts, and it is wrong if 2027 passes with French outages back within their normal range and no repeat of this summer's river transport problems.

Decision link: Strategic Implications [3](#) and [4](#).

Strategic Implications

Implication 1: Re-assess large connections as long contracts, not places in a queue

Whoever owns site strategy should re-cost every existing and planned large connection against the terms now being approved, before the next investment approval rather than after it. Oregon's terms run 10 to 30 years depending on size, make the customer pay 100% of local network upgrades, set minimum charges at 90% of the capacity it reserves and add a 1 cent/kWh surcharge above 100 MW ([Utility Dive](#), May 2026); twenty-three states now have a similar tariff, with seven more pending ([Columbia's Sabin Center](#), Jun 2026); and Texas requires a large user next to a power plant to be able to disconnect fully within thirty minutes ([White & Case](#), Aug 2026). These are balance-sheet commitments with exit costs, and they are being decided now, which is why this is a Decide rather than a Prepare. The reason to move before the next approval is that the six regional grid operators' responses to FERC, due by 17 August ([McGuireWoods](#), Jun 2026) will set the terms in the remaining markets while the approval is under way.

Action: by 31 October 2026, the site or business development lead produces a one-page summary of the terms of each live connection (contract length, minimum charge, share of upgrade costs and cost of leaving) and names the person who owns each.

DECIDE

Draws on Themes 1, 2 and 3.

Implication 2: Buy the delivery slot, not the technology

Capital planning should treat a place in the factory schedule as the scarce asset and decide whether to reserve one before the 2027 budget closes. GE Vernova's orders and reserved slots grew from 100 GW to 116 GW in a single quarter, and it expects at least 125 GW under contract by year end ([GE Vernova](#), Jul 2026), Mitsubishi is delivering first-quarter 2026 orders in 2028 to 2030 ([Utility Dive](#), Aug 2026), and transformers run to four years ([pv magazine USA](#), May 2026). The posture is Prepare rather than Decide because the right answer depends on how much guaranteed capacity the organisation needs, which it may not yet have worked out: what actually came online in the first half of 2026 was 11,458 MW of solar and 8,207 MW of batteries against 2,707 MW of gas ([Inside Climate News](#), Aug 2026), and for many patterns of demand that is the better stopgap. The decision to avoid is waiting for turbine prices to fall, at \$2,500 per kW and rising ([IEEFA](#), Apr 2026).

Action: by 15 December 2026, capital planning states how much guaranteed capacity it needs, by season and hour, for 2029 to 2032, and whether to reserve a delivery slot against it.

PREPARE

Draws on Themes 4 and 1.

Implication 3: Put a water and river-temperature test into valuation and fuel logistics

Asset owners, lenders and insurers should add a low-water case to the valuation of coal, gas and nuclear plants, and to any fuel contract that moves by barge, ahead of the next revaluation cycle rather than after the next summer. This season produced a full Cernavoda shutdown for only the second time in the plant's history ([Euronews](#), Aug 2026), a record 15.6% of French capacity missing ([Agence France-Presse, via Boursorama](#), Aug 2026), Alpine hydro at roughly half of normal ([Euronews](#), Aug 2026) and a three-quarters cut in inland coal barge loadings with the fuel itself in surplus at the coast ([Insurance Journal](#), Jul 2026). The posture is Prepare because no blackout followed and France's system operator was right that supply security required no particular vigilance ([RTE](#), Jun 2026): the risk is to how often plants can run, what that costs and what contracts say, not a failure of the whole system. That is exactly the kind of risk a valuation model absorbs quietly and a lending committee never sees.

Action: before the next annual revaluation, the risk team adds a six-week low-water case to every river-cooled plant and barge-supplied asset it holds, and reports the revenue at risk to the audit or risk committee.

PREPARE

Draws on Themes 5 and 3.

Implication 4: Track how each jurisdiction chooses to share the cost of the grid

Government affairs and regulatory teams should keep a current view of the tool used in each market, because the tools are diverging and each leaves the leftover cost with someone different. The United States is using state tariffs ([Columbia's Sabin Center](#), Jun 2026) alongside a voluntary federal pledge claimed to cover 80% of power delivered to homes and businesses ([the White House](#), Jul 2026); Germany is subsidising transmission charges ([Bundesnetzagentur](#), May 2026); the Commission proposed cutting network charges for specific consumer groups ([European Commission](#), updated Jul 2026); Spain would make access revocable ([Aragon Digital](#), Sep 2026). Monitor is the right posture because none has yet been tested by demand falling short, and what matters is not the announcement but the first tool that survives that test.

Action: from Q4 2026, the regulatory lead keeps a one-line-per-jurisdiction list of these tools, reviewed at each quarterly risk meeting, and flags any move from a voluntary promise to a binding tariff, or from a fine to the threat of disconnection.

MONITOR

Draws on Themes 3, 2 and 1.

Scenario Matrix 2030

LARGE LOADS' POWER: SHARED GRID → OWN SUPPLY

Build and Bill

Large loads connect to the shared grid on long contracts that make them pay for what they trigger, on the Oregon model ([Utility Dive](#), May 2026). The network grows on their money, and capacity auctions start clearing below their caps again as solar, storage and flexible demand arrive. Equipment stays scarce until the turbine makers' new factories open, so the first years are slow. Heat and low water remain the main surprise for the existing fleet.

Early indicators: Second and third states adopt Oregon-style terms; a capacity auction clears below its cap without a rule change; a large-load tariff survives a cancelled project without the cost passing to households.

Power Islands

Large loads build or contract their own generation, often gas on site, and pay their own way: the IEA sees 15 to 27 GW of on-site gas at data centres by 2030 ([IEA](#), Apr 2026). The grid is spared the cost but loses the flexibility and the customers it was counting on, and regulators respond with standby and exit charges. Turbine slots become the prize, and whoever reserved early wins.

Early indicators: Co-location approvals speed up after the Texas order; on-site gas announcements outpace grid connection requests; standby-charge filings rise.

The Blame Auction

The closest to today. Large loads connect to the shared grid, but the cost lands on household bills, because tariffs are announced and not tested and price caps are set politically ([Governor of Maryland](#), Jul 2026). The shortfall grows under the cap, emergency powers stand in for buying new supply,

HOUSEHOLDS PAY · SHARED GRID

nobody has agreed to defend.

Early indicators: A fourth consecutive capped auction; a state leaving a negotiated price ceiling; electricity bills becoming an election issue in more states.

Behind the Meter

Large loads supply themselves, but the network they leave behind is still paid for by everyone else. Household bills rise for reasons that have nothing to do with household use, the base that pays for the grid shrinks, and the politics sharpens. The contest moves from the terms of connecting to the

term **HOUSEHOLDS PAY · OWN SUPPLY** reputational risk for operators arrives before the regulatory one.

Early indicators: A utility reports falling regulated load while regional consumption rises; exit-fee proposals appear; states debate charging departing loads for network costs left behind.

Options: Connecting a large load (developers and data-centre

operators)

OPTION A · LEAST CHANGE

Take the standard terms

Accept the tariff and the queue on offer, and connect when the grid is ready.

Trade-off: No extra cost now. The slowest route, and the terms can change while you wait.

▣ **Fits scenarios:** Build and Bill; The Blame Auction

OPTION B · MIDDLE COURSE

Offer flexibility

Accept a connection that can be cut back when the grid is under stress, in return for connecting sooner.

Trade-off: Faster connection. Operations must cope with being curtailed.

▣ **Fits scenarios:** All four

OPTION C · BIGGEST SHIFT

Bring your own power

Build or contract generation and storage on site, and use the grid as backup.

Trade-off: Most control, most capital. Exposed to standby and exit charges.

▣ **Fits scenarios:** Power Islands; Behind the Meter

Options: Firm capacity (utilities, generators and large buyers)

OPTION A · LEAST CHANGE

Wait for prices to fall

Hold off on gas orders until the turbine makers' new factories open.

Trade-off: No capital at risk now. The supply gap stays open to the end of the decade.

▣ **Fits scenarios:** The Blame Auction

OPTION B · MIDDLE COURSE

Bridge with solar and storage

Build what can be delivered now and decide on firm capacity later.

Trade-off: Fast and available. Does not cover a still winter night.

▣ **Fits scenarios:** All four

OPTION C · BIGGEST SHIFT

Reserve a delivery slot

Pay now to secure a turbine or transformer slot in the late 2020s.

Trade-off: Locks in scarce equipment. Commits capital before the need is certain.

▣ **Fits scenarios:** Build and Bill; Power Islands

Options: Who pays for the grid (regulators and policymakers)

OPTION A · LEAST CHANGE

Cap the bill

Hold prices down with caps and subsidies, as Maryland, Pennsylvania and Germany have done.

Trade-off: Protects households now. Hides a shortfall that keeps growing underneath.

▣ **Fits scenarios:** The Blame Auction

OPTION B · MIDDLE COURSE

Make the large load pay

Long contracts with minimum charges and exit costs, on the Oregon model.

Trade-off: Protects households while developers stay solvent. Untested when demand disappoints.

▣ **Fits scenarios:** Build and Bill

OPTION C · BIGGEST SHIFT

Tie access to new supply

Connect large loads only if they bring new clean supply, and withdraw access if they do not, as Spain's draft decree proposes.

Trade-off: Adds supply alongside demand. The strongest tool and the hardest to enforce.

▣ **Fits scenarios:** Build and Bill; Power Islands

Options: Water and heat (asset owners, lenders and insurers)

OPTION A · LEAST CHANGE

OPTION B · MIDDLE COURSE

Treat it as weather

Keep current availability assumptions and handle outages as they come.

Trade-off: No cost now. The next dry summer surprises the model.

▣ **Fits scenarios:** None for long

Price it in

Add a six-week water case to valuations and write low-water clauses into fuel contracts.

Trade-off: Moderate effort. Shows the risk before a lender or insurer does.

▣ **Fits scenarios:** All four

OPTION C · BIGGEST SHIFT

Adapt the assets

Invest in cooling, fuel stocks on site and supply routes that do not depend on one river.

Trade-off: Costly. Keeps plants running when rivers run low.

▣ **Fits scenarios:** All four

What We Are Not Planning For

SYSTEMIC RELIABILITY

A drought-driven European supply failure this winter

The 2026 season took generation offline without taking the system down, and the operator best placed to judge said so in advance: France's system operator stated that security of electricity supply required no particular vigilance ([RTE](#), Jun 2026), and no shortfall followed. The Joint Research Centre's own comparison puts 9% of EU and UK territory at alert level in July against 18% at the 2022 peak ([the Commission's Joint Research Centre](#), Aug 2026), so this was a broader but less intense event than the one Europe already absorbed. The risk this briefing covers is how often plants can run and what that costs, which is why it sits inside Theme 5 rather than here.

↻ **Reinstated if** a member state issues a winter supply warning attributable to hydrology, or French nuclear outages exceed their 2022 peak.

JURISDICTION

A single US federal rule on who pays for the grid, overriding the states

It would resolve the messiest question in this cycle at a stroke, and there is no evidence it is coming. FERC explicitly left states in charge of retail sales, who qualifies and where plants are built when it issued its orders to the grid operators ([Utility Dive](#), Jun 2026), and chose regional rules over a national standard ([McGuireWoods](#), Jun 2026). The federal instrument actually deployed was a voluntary pledge ([the White House](#), Jul 2026), which is the opposite of overriding the states. Planning for a single national rule would mean ignoring the twenty-three different instruments that already exist ([Columbia's Sabin Center](#), Jun 2026).

↻ **Reinstated if** FERC starts writing rules on how retail customers share grid costs, or a federal bill on pricing for very large users gains committee support.

DEMAND

A collapse in data-centre electricity demand inside the planning horizon

The demand correction that would relieve every constraint in this briefing is not visible in the spending. Spending by the five largest cloud companies reached roughly \$490bn in the twelve months to May 2026 ([FactSet](#), Jul 2026), and ERCOT is planning against roughly 175,000 MW by 2032 ([The Texas Tribune](#), Jul 2026). What did change is how it is paid for, with new debt rising from 9% to 32% of that spending ([FactSet](#), Jul 2026). That is a real risk and it is tracked in Theme 3's counter-argument, but it is a risk to returns and to who pays for network built for demand that never came, not a near-term collapse in demand.

↺ **Reinstated if** the largest cloud companies cut their spending plans for two quarters running, or a regional grid operator reports a net fall in requests to connect very large users.

Questions For The Next Strategy Meeting

Five questions this cycle's evidence should put on the agenda.

QUESTION 1

If a capacity price at the ceiling the rules allow procured 317.1 MW against a 6,831.3 MW shortfall, what evidence would persuade us that the next auction will actually buy new supply rather than just decide who pays, and what do we do differently in the meantime?

QUESTION 2

Oregon's approved terms run to thirty years with minimum charges at 90% of contracted capacity. Which of our live connection plans would we still approve on those terms, and who signs that off?

QUESTION 3

We can reserve a turbine slot or keep our options open on technology, and with transformers taking four years we probably cannot do both. Which do we choose, and how much guaranteed capacity do we need?

QUESTION 4

If a large user that signed a tariff making it pay for its network goes bust in 2031, does the cost fall on its administrator or on household bills, and have we read the tariff in every jurisdiction where we are exposed?

QUESTION 5

Jellyfish and a 34-year low on the Danube took capacity off two European systems in the same season. How much revenue do we lose if low water limits a plant for six weeks rather than six days, and is that number anywhere in our current model?

Source Confidence Register

This briefing draws on 50 verified sources, gathered under a hard six-month window (publications from March 2026 onward), with 3 structural anchors outside that window.

Source tiers: **Tier 1**, governments, regulators and intergovernmental bodies. **Tier 2**, think-tanks, academic institutes, major consultancies and quality data providers. **Tier 3**, quality journalism and specialist trade press. **Tier 4**, vendor, company and practitioner sources, used only as directional corroboration.

Theme 1: Paying more did not buy more power

Source	Tier	Date	Key claim used
PJM Interconnection	Tier 1	Jul 2026	Cleared at the \$325.00/MW-day UCAP cap, procured 149,181.6 MW UCAP in total, 6,831.3 MW below the RTO Reliability Requirement, with only 317.1 MW UCAP of new generation.
Utility Dive	Tier 3	Jul 2026	Third consecutive cap-clearing auction, about 525 MW of new resources including 208 MW of uprates, 6.8 GW below the 20% reserve margin target.

Source	Tier	Date	Key claim used
New Jersey Board of Public Utilities	Tier 1	Aug 2026	New Jersey bills rose roughly 20% last year driven largely by PJM capacity costs, which account for 15 to 20% of a typical bill; the state proposes large new users bring their own generation or accept lower-priority service.
PJM Interconnection	Tier 1	May 2026	PJM names a credibility trap in which the scarcity prices needed to signal investment trigger intervention that undermines it, and sets out three reform paths including explicit rationing by geography or customer class.
Office of Governor Wes Moore (Maryland)	Tier 1	Jul 2026	Maryland states the 2028/2029 BRA cleared at the administrative cap of \$325/MW-day and that the cap prevented \$13.3bn of excess costs across the 13-state footprint and the District of Columbia.
Commonwealth of Pennsylvania, Office of the Governor	Tier 1	Jul 2026	Pennsylvania puts the settlement's cumulative effect at \$18.2bn, with roughly \$45bn projected across four years for 67 million people and about \$207 per household in the 2028/2029 delivery year.
U.S. Energy Information Administration	Tier 1	Aug 2026	ERCOT set an hourly peak of 91.1 GW on 22 July 2026, 6% above the previous record of 85.5 GW from 10 August 2023, met primarily by natural gas at 48% and solar at 32%.
The Texas Tribune	Tier 3	Jul 2026	ERCOT projects roughly 175,000 MW of demand by 2032 against the 91,308 MW record of July 2026; the generation queue holds about 460,000 MW of interest, around 70% of it solar and storage and about 78,000 MW gas; at the moment the 22 July record was set the grid held more than 20,000 MW of buffer between demand and committed generation capacity.

Theme 2: The regulator rewrote the rules for every large load

Source	Tier	Date	Key claim used
McGuireWoods	Tier 2	Jun 2026	FERC ordered all six RTOs and their transmission owners on 18 June 2026 to justify or revise large-load interconnection tariffs, with responses due 17 August 2026 across five reform categories.
White & Case	Tier 2	Aug 2026	On 24 July 2026 the PUCT ruled that emergency curtailment of co-located large loads is not limited by the paired generator's capacity: the load must be able to disconnect fully within 30 minutes.
Ofgem	Tier 1	Jul 2026	Contracted demand offers rose from 41 GW to 125 GW in under a year, around 80 GW of it data centres; Ofgem proposes a commitment fee and progress milestones to clear speculative applications.
North American Electric Reliability Corporation	Tier 1	May 2026	Resources rose by over 58.5 GW and peak demand by over 11 GW since 2025; NERC flags voltage-sensitive load reduction events of roughly 1,500 MW in both the Eastern and ERCOT interconnections.
Herbert Smith Freehills Kramer	Tier 3	Jul 2026	Ofgem and DESNZ propose reforming demand-side connections with entry criteria, deposits or progression fees for data centres, and post-offer milestones covering customer acquisition and long-lead procurement.
Utility Dive	Tier 3	Jun 2026	FERC gave the six RTOs 60 days to respond, extendable to 150, required evaluation of alternative transmission technologies before conventional upgrades, and preserved state authority over retail sales and siting.

Source	Tier	Date	Key claim used
Federal Energy Regulatory Commission	Tier 1	Evergreen	The docket covers interconnection of loads generally above 20 MW and seeks input on expedited study timelines, cost allocation and reliability evaluation; the Secretary of Energy directed FERC to consider the ANOPR on 23 October 2025.
Agency for Natural Resources and Energy, Ministry of Economy, Trade and Industry (Japan) <i>(in Japanese)</i>	Tier 1	May 2026	From April 2026 Japanese data centre operators must report electricity use, PUE and energy intensity, publish targets annually by 31 March, reach PUE of 1.4 or below by fiscal 2030, and 1.3 or below for centres built from fiscal 2029.
International Energy Agency	Tier 1	Apr 2026	The IEA's updated projections see data centre electricity consumption roughly doubling from 485 TWh in 2025 to 950 TWh in 2030, around 3% of global electricity demand, and around 15 to 27 GW of onsite natural gas may power data centres by 2030, mostly in the United States.

Theme 3: Who pays for the grid became the live political question

Source	Tier	Date	Key claim used
Bundesnetzagentur <i>(in German)</i>	Tier 1	May 2026	The AgNES reform shifts consumers above 100,000 kWh a year from performance to capacity pricing, introduces capacity charges on generators of EUR 4 to 7 per kW a year, and raises base fees for prosumers.
Smart Electric Power Alliance	Tier 2	Apr 2026	77 approved and proposed large-load tariffs across 60 utilities as of 31 March 2026; state regulators approved 29 in 2025 alone against 14 in the whole of 2018 to 2024.
The White House	Tier 1	Jul 2026	The expanded pledge covers over 200 utilities, data centre developers, cooperatives and states, and claims to cover 80% of all power delivered to US homes and businesses.

Source	Tier	Date	Key claim used
Utility Dive	Tier 3	May 2026	Oregon approved PGE's Schedule 96 on 7 May 2026: contracts of 10 to 30 years by size, 100% of distribution upgrade costs on the customer, minimum demand charges at 90% of contracted capacity, and a 1 cent/kWh surcharge above 100 MW.
Sabin Center for Climate Change Law, Columbia Law School	Tier 2	Jun 2026	As of May 2026, 23 states have at least one large-load tariff in place and seven more have proposals pending; Virginia's takes effect 1 January 2027.
Sierra Club	Tier 3	Jul 2026	Cites an independent analysis putting the rise in annual customer costs in PJM at \$12.5bn since June 2025, from \$2.2bn in 2024 to \$14.7bn the following year, with 67 million customers paying more even under the negotiated ceiling.
Aragon Digital (<i>in Spanish</i>)	Tier 3	Sep 2026	Spain's draft decree would require data centres to source 80% of their energy from new renewable capacity through self-consumption or long-term contracts, with escalating charges on the electricity bill for failure and loss of network access for repeated breach; over 600 responses were filed in consultation.
European Commission, Directorate-General for Energy	Tier 1	Evergreen	The Commission adopted guidance on network charges in December 2025 as part of the Grids Package, and in July 2026 proposed reducing network charges for certain consumer groups and taxes for energy-intensive businesses alongside faster smart-meter deployment.

Theme 4: You cannot buy a turbine before 2030

Source	Tier	Date	Key claim used
POWER Magazine	Tier 3	Sep 2026	GE Vernova reported 116 GW of gas turbine capacity under contract at the end of Q2 2026, up from 100 GW three months earlier, and is mostly sold out through 2030.

Source	Tier	Date	Key claim used
Utility Dive	Tier 3	Aug 2026	Siemens Energy's gas services backlog reached 69 GW, with lead times of three years or more and expansion plans adding 15 large units in 2027 and 20 medium units by 2028.
pv magazine USA	Tier 3	May 2026	High-capacity transformer lead times extend to four years, with generator step-up transformer demand up 274% between 2019 and 2025 against constrained grain-oriented electrical steel and copper supply.
Utility Dive	Tier 3	Aug 2026	Mitsubishi's large-frame backlog reached 35 GW, up from 23 GW a year earlier; orders booked in Q1 2026 are scheduled for delivery between 2028 and 2030, and the company is doubling manufacturing capacity against 2024 levels.
Institute for Energy Economics and Financial Analysis	Tier 2	Apr 2026	The newest combined-cycle proposals price at least \$2,500 per kW installed, or \$2.5bn for a 1,000 MW unit, roughly triple early-2020s costs, pushing implied power prices toward \$107/MWh against earlier \$45 to \$75/MWh estimates.
Inside Climate News	Tier 3	Aug 2026	Of 368 utility-scale plants online in the US in the first half of 2026, solar added 11,458 MW and batteries 8,207 MW, against 2,707 MW of natural gas across 46 projects.
Utility Dive	Tier 3	Aug 2026	On Q2 2026 calls the three major gas turbine makers were reported with backlogs from 35 to 116 GW, and Duke's CEO said the company is deploying more than \$1bn a month to meet record demand.
GE Vernova	Tier 2	Jul 2026	Gas Power equipment backlog and slot reservation agreements grew from 100 GW to 116 GW in the quarter, with at least 125 GW anticipated under contract by year-end 2026 and annual output planned at 20 GW in Q3 2026, 24 GW in 2028 and 30 GW by 2030.

Source	Tier	Date	Key claim used
Utility Dive, reporting Wood Mackenzie	Tier 3	Apr 2026	Wood Mackenzie projects gas turbine prices rising to \$600/kW by the end of 2027, a 195% increase since 2019, with turbine demand peaking in 2026 as developers try to secure equipment for 63 GW of gas capacity additions from 2026 to 2030.

Theme 5: Heat and low water stopped being a weather story

Source	Tier	Date	Key claim used
Selectra (in French)	Tier 3	Aug 2026	13 of 57 French reactors affected, eight fully stopped and five at reduced power, 20.4% of capacity unavailable; Romania lost both Cernavoda units on low Danube levels; EDF climate losses around 0.3% a year could reach 1.4% by 2035.
RTE (in French)	Tier 1	Jun 2026	The system operator judged supply security to require no particular vigilance during the heatwave and expected generation to cover demand.
Agence France-Presse (via Boursorama) (in French)	Tier 3	Aug 2026	EDF nuclear output fell about 3% in June and 6% in July, with a record 15.6% of capacity missing on one Sunday; three Gravelines reactors were shut by tonnes of jellyfish in the sea-water intakes.
Euronews	Tier 3	Aug 2026	Both Cernavoda units were offline on 13 August 2026; the plant normally supplies a fifth of Romania's electricity and had been shut for drought only once before, in 2003.
Mediafax (in Romanian)	Tier 3	Jul 2026	Apele Romane put the Danube flow at Romania's border at 1,650 cubic metres per second in late July 2026 against a July average of 4,750, and forecast about 1,500 by early August against a 1985 record minimum of 1,400.

Source	Tier	Date	Key claim used
Insurance Journal	Tier 3	Jul 2026	Kaub fell to 28 centimetres in late July 2026, the lowest since 2018, with a forecast of 24 centimetres, the lowest in records back to 1990; ARA coal inventories rose about 75% since spring while barge loadings inland were cut by three-quarters.
Joint Research Centre, European Commission	Tier 1	Aug 2026	The European Drought Observatory recorded half of the EU and the UK under drought at some level and 9% of the territory at alert level in July 2026, against 18% at alert at the 2022 peak in mid-August and 6% in late August 2018.
Euronews	Tier 3	Aug 2026	July 2026 hydro output ran at 51% of normal in Austria, 48% in Switzerland, 63% in Germany and 68% in France, with EU gas generation up 8% last year against 2024 as hydro fell 12%.
MIT Technology Review	Tier 3	Jun 2026	Extreme heat is forcing European power plants to cut output while pushing up demand for cooling; globally, energy use for cooling is set to double by 2050 relative to 2023 levels.

Continuity Ledger

Source	Tier	Date	Key claim used
PJM Interconnection	Tier 1	Jul 2026	PJM hit an all-time peak of 168,158 MW between 5 and 6 p.m. on 2 July 2026, past the 165,563 MW record of 2 August 2006, with demand response of about 6,113 MW on 2 July and 5,037 MW on 3 July; the backup-generator curtailment action was authorised but never issued.
Utility Dive	Tier 3	May 2026	The DOE order of 18 May 2026 let PJM curtail large loads with backup generation as a last resort before rolling blackouts; PJM expected under 5,800 MW of reserves at the 18 May peak with more than 40 GW offline for planned outages.

Source	Tier	Date	Key claim used
FactSet	Tier 2	Jul 2026	Aggregate capex across the five hyperscalers rose from \$95bn of investing cashflows in FY20 to around \$490bn in the twelve months to May 2026, with incremental debt as a share of capex up from 9% in FY24 to 32% in the year before June 2026 and total debt near \$700bn.
Insurance Business	Tier 3	Jul 2026	Global insured natural catastrophe losses were \$46bn in H1 2026, the lowest first half since 2019 and 28% below the \$64bn ten-year average, with only 11 events above \$1bn against a ten-year average of 16; property reinsurance rates fell 12% at 1 January and 16% at the July midyear renewal.
Intelligent Insurer	Tier 3	Apr 2026	The FAIR Plan added 15,779 policies in Q1 2026, a 2.4% gain, lifting total exposure by \$27bn or 3.7% in the quarter and 25% year on year.
Data Center Knowledge	Tier 3	Jul 2026	A Virginia DEQ study found no location east of Interstate 95 could support a hypothetical three million gallon per day industrial groundwater withdrawal under current permitting standards, with sustainable limits ranging from 360,000 gallons per day in Northumberland County to under 30,000 in Henrico and Sussex.
Bank of England	Tier 1	Evergreen	The 2026 Dynamic General Insurance Stress Test closed its live simulation phase in May 2026; the Bank is analysing firm submissions and states that findings will be published at an aggregate level later in 2026.

Claim-fidelity self-disclosure

This appendix records every claim in the briefing that goes beyond what a cited source literally states, so a reader can see where the analyst's judgment enters.

Cycle-over-cycle continuity claims

Every statement in this briefing about what a previous edition said is checked against that edition's own record, not against memory or paraphrase. Eight appear, all inside the Continuity ledger. Seven grade the seven milestones the 16 July edition set for this cycle: the Q2 capex prints, the Atlantic season against the soft-pricing bet, whether any grid executed data-centre curtailment, the California FAIR Plan trajectory, data-centre water-permit denials, the Bank of England stress-test results, and the PJM 2028/2029 clearing price. The eighth is a correction rather than a grade. The July edition recorded the 2 July PJM peak as roughly 162.7 GW and concluded that the 2006 record had held; PJM's own account puts the peak at 168,158 MW, above the 165,563 MW record. That reading is withdrawn here rather than quietly restated, because it was load-bearing for July's argument that the grid passed the summer without breaking anything.

Superlatives used

Every "first", "only", "largest" and "record" in the briefing was checked against the source behind it. Most turn out to be calendar or counting words rather than claims ("first quarter", "first half of 2026", "only 9% of territory"). Five are genuine superlatives and each is carried by a source using the same word for the same referent: "only the second time in its history" for the Cernavoda shutdown, and "just once before, in 2003", are Euronews's characterisation; "record 15.6%" of French capacity missing is AFP's; "all-time peak" for the 2 July PJM record is PJM's own; "lowest since 2018" for the Rhine at Kaub is Insurance Journal's. One is analyst characterisation with no source behind it, kept because it is framing rather than a claim, and disclosed here: "the clearest measure of the gap" for GE Vernova's own figures. Three further superlatives were removed at draft rather than defended: an earlier sit-up line claiming the auction cleared at "the highest price on record" was factually wrong, because the 2027/2028 auction cleared higher at \$333.44/MW-day, and now reads "the highest price the rules allow"; "two of Europe's largest rivers" is now "the Rhine and the Danube"; and Japan's PUE floor is described as "a hard performance number" rather than "the first".

Analytical synthesis claims

Four judgments read across sources and none is stated by any single source. First, that the binding constraint is delivery rather than willingness to pay: this draws on the PJM auction report, PJM's market-design paper, the three OEM order books and the EIA and Texas Tribune ERCOT figures, and is stated in the Executive Synthesis as the analysis's own reading. Second, that a price cap which holds the bill down while the shortfall widens manages bills rather than buying supply: this reads the Maryland and Pennsylvania releases against the auction report, appears in the Options and in The Blame Auction scenario, and both governments would contest the characterisation. Third, that tariffs for large users move cost without moving the risk that demand falls short: this combines the Sabin Center's state count, the Oregon terms and the FactSet funding data, and is framed in the text as "on the available evidence". Fourth, that river flow and

temperature have become scheduling inputs rather than tail risks: this reads the Romanian, French, Alpine and Rhine evidence together and is stated in Theme 5's opening paragraph. The counter-argument in each theme is the disconfirming case for that judgment, not a rival frame.

Two figures in this briefing are analyst constructions with no source entry of their own. The bar chart restates figures from the PJM auction report to scale. The convergence diagram is drawn from the cross-impact grid in the same topic's Change Tracker and re-sourced against this cycle's evidence; its compound-exposure statement is the analyst's, not a source's.

Analyst judgements and added sources. The theme weights, the "who decides" and "room to act" lines, the Options and the new scenario axes are the analysis's proposals for discussion, not findings. The scenario matrix was redrawn on 24 September 2026 with a 2030 horizon; the first version asked how fast new supply arrives and whether cost allocation settles. Three forward-looking sources were added for the Outlook to 2030 blocks: the IEA's updated data-centre projections ([IEA](#), Apr 2026), Wood Mackenzie's turbine price forecast as reported by Utility Dive ([Wood Mackenzie, reported by Utility Dive](#), Apr 2026), and MIT Technology Review's report on heat and power plants ([MIT Technology Review](#), Jun 2026). All three carry forecasts and are reported as such; the second reaches us through journalism rather than the original report. In the same revision the briefing's wording was made plainer for readers outside the sector: technical terms are explained where they appear, and GE Vernova's quarterly statement, quoted word for word in the first published version, is now summarised in plain words. Its exact wording is kept in the evidence record, and no claim or figure was changed.

Abbreviations

The abbreviations used in this briefing, in alphabetical order.

AFP: Agence France-Presse, the French news agency

ANOPR: Advance notice of proposed rulemaking, an early step in US federal rule-making

ARA: Amsterdam, Rotterdam and Antwerp, Europe's main coal import ports

BRA: Base Residual Auction, PJM's main auction for power capacity

DESNZ: Department for Energy Security and Net Zero (UK government)

DOE: US Department of Energy

EDF: Électricité de France, France's main electricity generator

ERCOT: Electric Reliability Council of Texas, which runs the grid for most of Texas

FERC: Federal Energy Regulatory Commission, the US federal energy regulator

GW: Gigawatt, one thousand megawatts

IEA: International Energy Agency

IEEFA: Institute for Energy Economics and Financial Analysis

kW: Kilowatt, one thousand watts

kWh: Kilowatt-hour, the unit on a household electricity bill

MW: Megawatt, one thousand kilowatts

MW-day: Megawatt-day, the unit in which capacity is priced: one megawatt available for one day

MWh: Megawatt-hour, one thousand kilowatt-hours

NERC: North American Electric Reliability Corporation, which sets reliability standards for the North American grid

OEM: Original equipment manufacturer; here, the turbine makers

PGE: Portland General Electric, an Oregon utility

PJM: PJM Interconnection, which runs the grid and power markets for 13 US states and Washington DC

PUCT: Public Utility Commission of Texas

PUE: Power usage effectiveness: a data centre's total electricity use divided by the electricity its computers use; 1.0 is perfect

RTE: Réseau de Transport d'Électricité, France's grid operator

RTO: Regional transmission organisation, which runs the grid and power markets across several US states

TWh: Terawatt-hour, one million megawatt-hours

UCAP: Unforced capacity, the share of a plant's capacity that PJM counts as reliably available

This Decision Intelligence briefing is the quarterly synthesis of a Shaping Tomorrow intelligence cycle on **Energy, Infrastructure & Climate Resilience**. It sits on top of weekly Signal Scans that identify emerging developments and weak signals, and monthly Change Trackers that assess what is materially shifting in momentum. Cycles may also close with Action Triggers, an end-of-cycle synthesis that reads across all four formats and surfaces patterns no single deliverable carries on its own. Continuous intelligence on this topic is available as a Single-Topic or Multi-Topic programme.

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